

Text as Data, Homework 1

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Packages

Uploading Files

Tidy text & word frequencies

The `unnest_tokens` function

```
full_text_tidy <- eo_data |>
  select(document_number, title, publication_date, signing_date, full_text) |>
  unnest_tokens(word, full_text)

View(full_text_tidy)
```

Removing non-alphanumeric characters

```
full_text_tidy <- full_text_tidy %>%
  mutate(word = str_extract(word, "[a-z']+"))
```

Stop words

```
full_text_tidy <- full_text_tidy %>%
  anti_join(stop_words)
```

Joining with ``by = join_by(word)``

```
full_text_tidy
```

```
# A tibble: 173,215 x 5
```

	document_number	title	publication_date	signing_date	word
	<chr>	<chr>	<date>	<date>	<chr>
1	2016-30101	Relating to the Implemen~	2016-12-13	2016-12-08	title
2	2016-30101	Relating to the Implemen~	2016-12-13	2016-12-08	<NA>
3	2016-30101	Relating to the Implemen~	2016-12-13	2016-12-08	pres~
4	2016-30101	Relating to the Implemen~	2016-12-13	2016-12-08	exec~
5	2016-30101	Relating to the Implemen~	2016-12-13	2016-12-08	<NA>
6	2016-30101	Relating to the Implemen~	2016-12-13	2016-12-08	dece~
7	2016-30101	Relating to the Implemen~	2016-12-13	2016-12-08	<NA>
8	2016-30101	Relating to the Implemen~	2016-12-13	2016-12-08	rela~
9	2016-30101	Relating to the Implemen~	2016-12-13	2016-12-08	impl~
10	2016-30101	Relating to the Implemen~	2016-12-13	2016-12-08	conv~

```
# i 173,205 more rows
```

```
meaningless_words <- tibble(word=c("thereof", "section", "sec", "ii", "iii", "iv", "vi", "vi  
full_text_tidy <- full_text_tidy %>%  
  anti_join(meaningless_words)
```

```
Joining with `by = join_by(word)`
```

Analysing word frequencies

```
fulltext_freq <- full_text_tidy %>%  
  filter(!is.na(word)) %>%  
  count(word, sort=T)  
  
head(fulltext_freq, 20)
```

```
# A tibble: 20 x 2
```

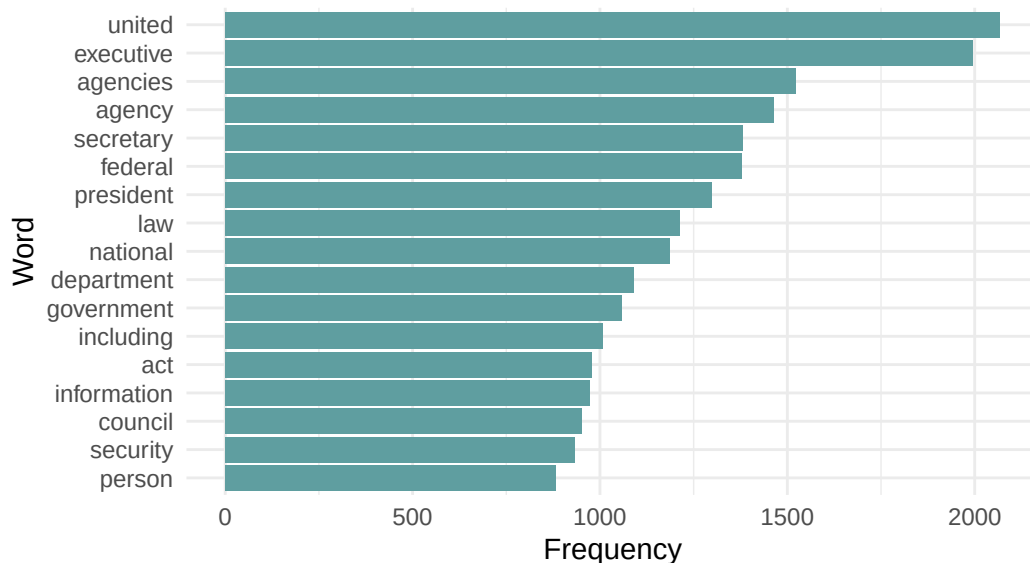
	word	n
	<chr>	<int>
1	united	2068
2	executive	1994
3	agencies	1523
4	agency	1463

5	secretary	1381
6	federal	1379
7	president	1298
8	law	1214
9	national	1187
10	department	1090
11	government	1058
12	including	1008
13	act	979
14	information	974
15	council	951
16	security	933
17	person	883
18	director	775
19	office	747
20	authority	726

Plotting word frequencies with bar graphs

```
fulltext_freq %>%
  group_by(word) %>%
  summarise(n = sum(n)) %>%
  filter(n>800) %>%
  ggplot(aes(x=n, y=reorder(word, n), fill=n)) +
  geom_col(fill = "cadetblue")+
  labs(
    x= "Frequency",
    y= "Word",
    title = str_wrap("Most Frequent Bigrams in President Obama's Executive Orders", width = 80)
  ) +
  theme_minimal()
```

Most Frequent Bigrams in President Obama's Executive Orders



TF-IDF

```
fulltext_idf <- full_text_tidy %>%
  filter(word != "na") %>%
  count(title, word, sort = TRUE) %>%
  bind_tf_idf(word, title, n)

fulltext_idf %>%
  select(title, word, tf_idf) %>%
  arrange(desc(tf_idf))
```

A tibble: 66,736 x 3

	title	word	tf_idf
	<chr>	<chr>	<dbl>
1	Accelerating Improvements in HIV Prevention and Care in the Uni~	hiv	0.338
2	Interagency Group on Insular Areas	igia	0.321
3	Correction	page	0.307
4	Advancing the Global Health Security Agenda To Achieve a World ~	ghsa	0.296
5	Amendment to Executive Order 11155, Awards for Special Capabili~	scho~	0.264
6	Chesapeake Bay Protection and Restoration	ches~	0.258
7	President's Council of Advisors on Science and Technology	pcast	0.257

```

8 Establishment of the President's Economic Recovery Advisory Board perab 0.251
9 Hostage Recovery Activities host~ 0.245
10 Amendments to Executive Order 12777 liab~ 0.241
# i 66,726 more rows

```

```

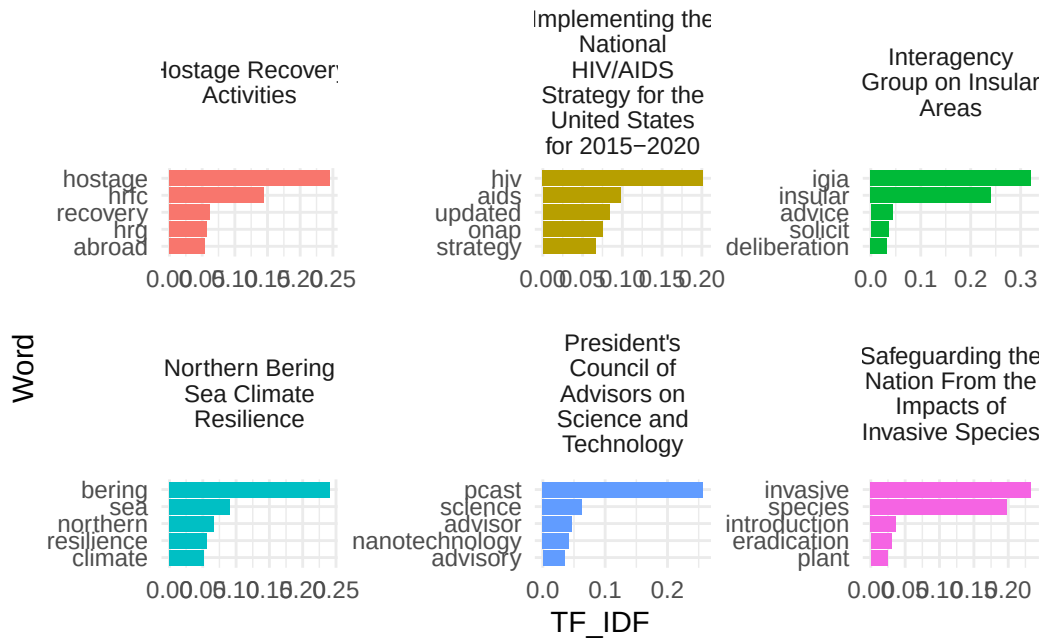
#| fig-width: 14
#| fig-height: 14

```

```

fulltext_idf %>%
  filter(title %in% c("Interagency Group on Insular Areas", "President's Council of Advisors on Science and Technology", "National HIV/AIDS Strategy for the United States for 2015-2020", "Safeguarding the Nation From the Impacts of Invasive Species", "Implementing the National HIV/AIDS Strategy for the United States for 2015-2020", "Hostage Recovery Activities", "Northern Bering Sea Climate Resilience")) %>%
  group_by(title) %>%
  arrange(desc(tf_idf)) %>%
  top_n(5, tf_idf) %>%
  ggplot(aes(x=tf_idf, y=reorder(word, tf_idf), fill = title)) +
  geom_col(show.legend = F) +
  labs(x="TF_IDF", y= "Word") +
  facet_wrap(~title, scales = "free", labeller = label_wrap_gen(width = 17)) +
  theme_minimal()

```



Analyzing n-grams

```
eo_structured <- eo_data %>%
  select(document_number, title, signing_date, full_text) %>%
  filter(!is.na(full_text)) %>%
  mutate(
    sections = str_split(full_text, "(?=Sec\\.\\.\\.s*\\.d|Section\\s+\\.d)")
  ) %>%
  unnest(sections) %>%
  mutate(
    section_text = str_trim(sections),
    section_num = row_number()
  ) %>%
  filter(section_text != "")
```

Bigrams

```
eo_bigrams <- eo_data %>%
  select(document_number, title, signing_date, full_text) %>%
  filter(!is.na(full_text)) %>%
  filter(year(signing_date) == 2015) %>%
  unnest_tokens(bigram, full_text, token = "ngrams", n = 2)
```

Counting n-grams

```
eo_bigrams %>%
  count(bigram) %>%
  arrange(desc(n))
```

```
# A tibble: 15,910 x 2
  bigram          n
  <chr>         <int>
1 of the         571
2 this order     382
3 of this        238
4 to the         219
5 united states  208
```

```

6 the united      179
7 executive order  135
8 by the          118
9 with the        116
10 shall be       99
# i 15,900 more rows

```

```

eo_bigrams %>%
  drop_na(bigram)

```

```

# A tibble: 37,665 x 4
  document_number title                signing_date bigram
  <chr>            <chr>                <date>        <chr>
1 2015-05677       Blocking Property and Suspending Entry o~ 2015-03-08    title~
2 2015-05677       Blocking Property and Suspending Entry o~ 2015-03-08    3 the
3 2015-05677       Blocking Property and Suspending Entry o~ 2015-03-08    the p~
4 2015-05677       Blocking Property and Suspending Entry o~ 2015-03-08    presi~
5 2015-05677       Blocking Property and Suspending Entry o~ 2015-03-08    execu~
6 2015-05677       Blocking Property and Suspending Entry o~ 2015-03-08    order~
7 2015-05677       Blocking Property and Suspending Entry o~ 2015-03-08    13692~
8 2015-05677       Blocking Property and Suspending Entry o~ 2015-03-08    of ma~
9 2015-05677       Blocking Property and Suspending Entry o~ 2015-03-08    march~
10 2015-05677       Blocking Property and Suspending Entry o~ 2015-03-08    8 201~
# i 37,655 more rows

```

Removing stopwords from n-grams

```

eo_bigrams <- eo_bigrams %>%
  separate(col = bigram,
    into = c("word1", "word2"),
    sep = " ",
    remove = F)

eo_bigrams_stop <- eo_bigrams %>%
  filter(!word1 %in% stop_words$word & !word2 %in% stop_words$word) %>%
  filter(!str_detect(bigram, "^sec(tion)?\\s+\\d")) %>%
  filter(!str_detect(bigram, "^title\\s+\\d")) %>%
  filter(!str_detect(word1, "^\\d+$") & !str_detect(word2, "^\\d+$")) %>%
  filter(!str_detect(bigram, "^fr\\s+"))

```

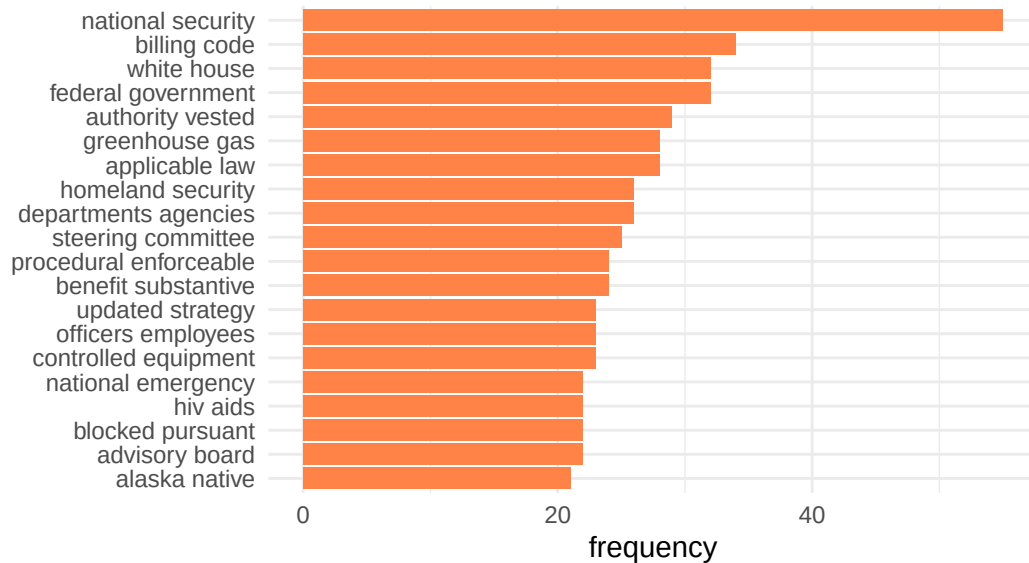
```
eo_bigrams_stop %>%
  count(bigram, sort = T)
```

```
# A tibble: 4,283 x 2
  bigram          n
  <chr>         <int>
1 national security    55
2 billing code         34
3 federal government   32
4 white house         32
5 authority vested    29
6 applicable law       28
7 greenhouse gas      28
8 departments agencies 26
9 homeland security    26
10 steering committee  25
# i 4,273 more rows
```

Plot bigram frequencies

```
eo_bigrams_stop %>%
  count(bigram, sort = T) %>%
  filter(n>20) %>%
  ggplot(aes(x=reorder(bigram, n),
                    y = n,
                    fill = n)) +
  geom_col(fill = "sienna1") +
  labs(x = NULL, y = "frequency",
        title = str_wrap("Most Frequent Bigrams in President Obama's 2015 Executive Orders", w
  coord_flip() +
  theme_minimal()
```


Most Frequent Bigrams in President Obama's 2015 Executive Orders



Filtering n-grams

```
eo_bigrams_stop %>%
  filter(str_detect(bigram, "government")) %>%
  distinct(bigram)
```

```
# A tibble: 57 x 1
  bigram
  <chr>
1 antigovernment protests
2 antigovernment protestors
3 term government
4 government consistent
5 government effort
6 government industry
7 federal government
8 government approach
9 government strategy
10 representing government
# i 47 more rows
```

TF_IDF with n-grams

```
eo_bigrams_tfidf <- eo_bigrams_stop %>%
  count(document_number, bigram) %>%
  bind_tf_idf(bigram, document_number, n)

eo_bigrams_tfidf %>%
  arrange(desc(tf_idf))
```

A tibble: 5,294 x 6

	document_number	bigram	n	tf	idf	tf_idf
	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>
1	2015-15828	presidential scholars	2	0.118	3.37	0.396
2	2015-01255	controlled equipment	23	0.0888	3.37	0.299
3	2015-29498	emergency board	7	0.109	2.67	0.292
4	2015-31749	thursday december	3	0.0811	3.37	0.273
5	2015-16334	advisory board	12	0.12	2.27	0.272
6	2015-25744	security medal	7	0.0805	3.37	0.271
7	2015-32582	attached hereto	6	0.08	3.37	0.269
8	2015-15495	courts martial	3	0.0769	3.37	0.259
9	2015-15495	martial united	3	0.0769	3.37	0.259
10	2015-18292	u.s trade	9	0.0744	3.37	0.250

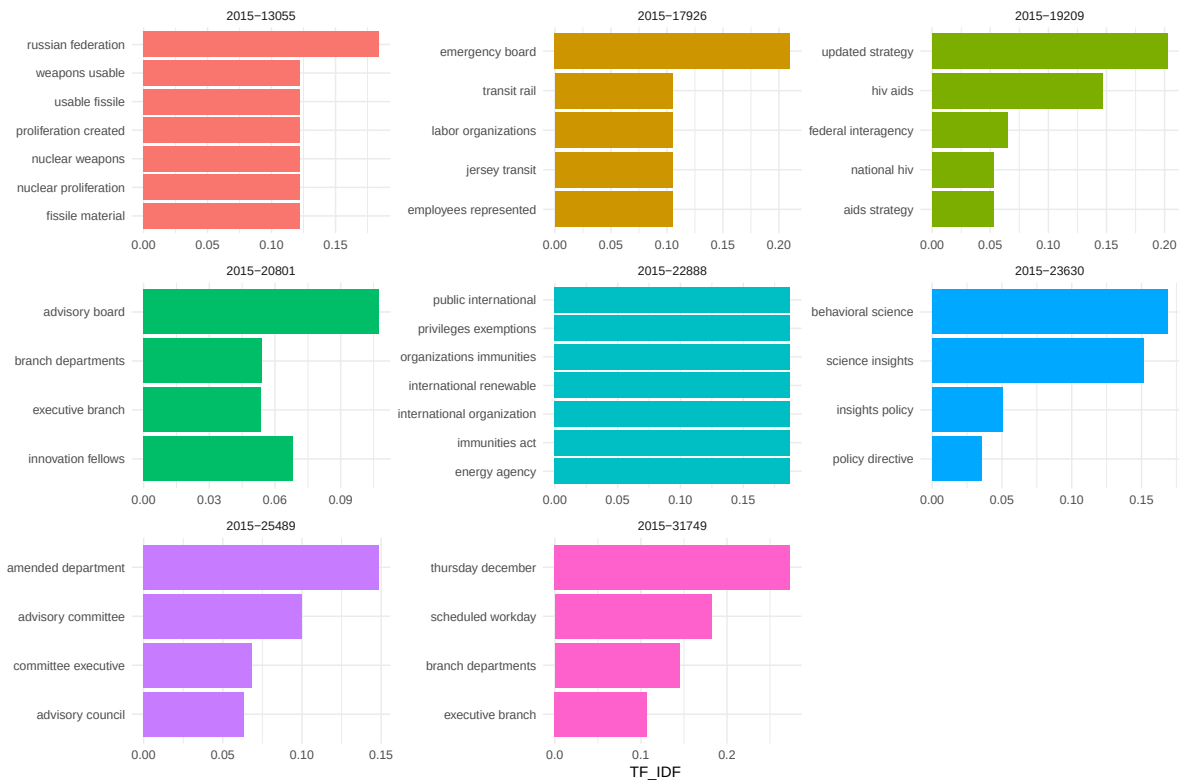
i 5,284 more rows

TF_IDF Plot

```
set.seed(148)
selected_docs <- sample(unique(eo_bigrams_tfidf$document_number), 8)

eo_bigrams_tfidf %>%
  filter(document_number %in% selected_docs) %>%
  group_by(document_number) %>%
  slice_max(tf_idf, n = 4) %>%
  ungroup() %>%
  ggplot() +
  aes(x = tf_idf,
      y = fct_reorder(bigram, tf_idf),
      fill = document_number) +
  geom_col(show.legend = F) +
```

```
facet_wrap(~document_number, scales = "free") +
labs(x = "TF_IDF", y = NULL) +
theme_minimal()
```



Creating and visualizing a bigram network

```
eo2015_graph <- eo_bigrams_stop %>%
  count(word1, word2) %>%
  filter(n > 15) %>%
  graph_from_data_frame()

set.seed(1049)

a <- grid::arrow(type = "closed", length = unit(.13, "inches"))

ggraph(eo2015_graph, layout = "fr") +
  geom_edge_link(aes(edge_alpha = n),
```


The word frequency analysis highlights the highly procedural nature of executive orders. Terms such as “federal,” “agency,” “secretary,” and “authority” appear frequently across the corpus, reflecting the fact that executive orders are primarily directives to government agencies rather than public-facing political statements. While useful, simple word counts reveal relatively little about the specific policy content of individual orders. Something I found particularly interesting in the analysis was that there was less of an overlap between the most common single words and the bigrams than I anticipated. I was not expecting the word section, because I filtered out all the section combinations, as they were acting as stop words as opposed to content, but I was expecting a greater overlap in some of the other words. For the bigrams, I was surprised that “United States” was not a combination that appeared over 20 times throughout the 2015 EOs, especially considering “united” was the second most used word in all of the EOs. I’m assuming that reference to the nation was done so in a broader way, but it was still interesting to note.

A limitation that I noticed was that when filtering for 2015 EOs, it was more difficult to sort words by their TF-IDF. Since there were so many fewer words compared to the entire dataset, there was less variety in the TF-IDF scores, which made graphing a bit more complicated. Because of this, I am inclined to prioritize larger sets of data because it gives more variety and more concrete analysis. Of course, large datasets have their issues, such as having to manually wade through some of the data, but I found in this assignment that working with the entire copra was easier than just the 2015 data.

LLM Statement: When scraping the .txt files from the dataset, I was running into issues with Federal Government request access. To figure out how to remedy this, I copy and pasted the request access note and asked ClaudeAI how I could get around this problem. ClaudeAI suggested a for loop and the use of the federal registrar website, as opposed to the dataset itself. ClaudeAI also gave me the code used from lines 25 to 56 to get around the issue. I looked over the code and when I ran it, it worked as intended. I also used ClaudeAI for lines 60 to 64, as I could not figure out how to combine my .txt files into one document to use for tidytext analysis. For this, I told Claude AI my problem and asked for a few different ways to go about combining the files. I decided on the one I ended up using because it was the most straightforward and would be the easiest to use for the purposes of the assignment.